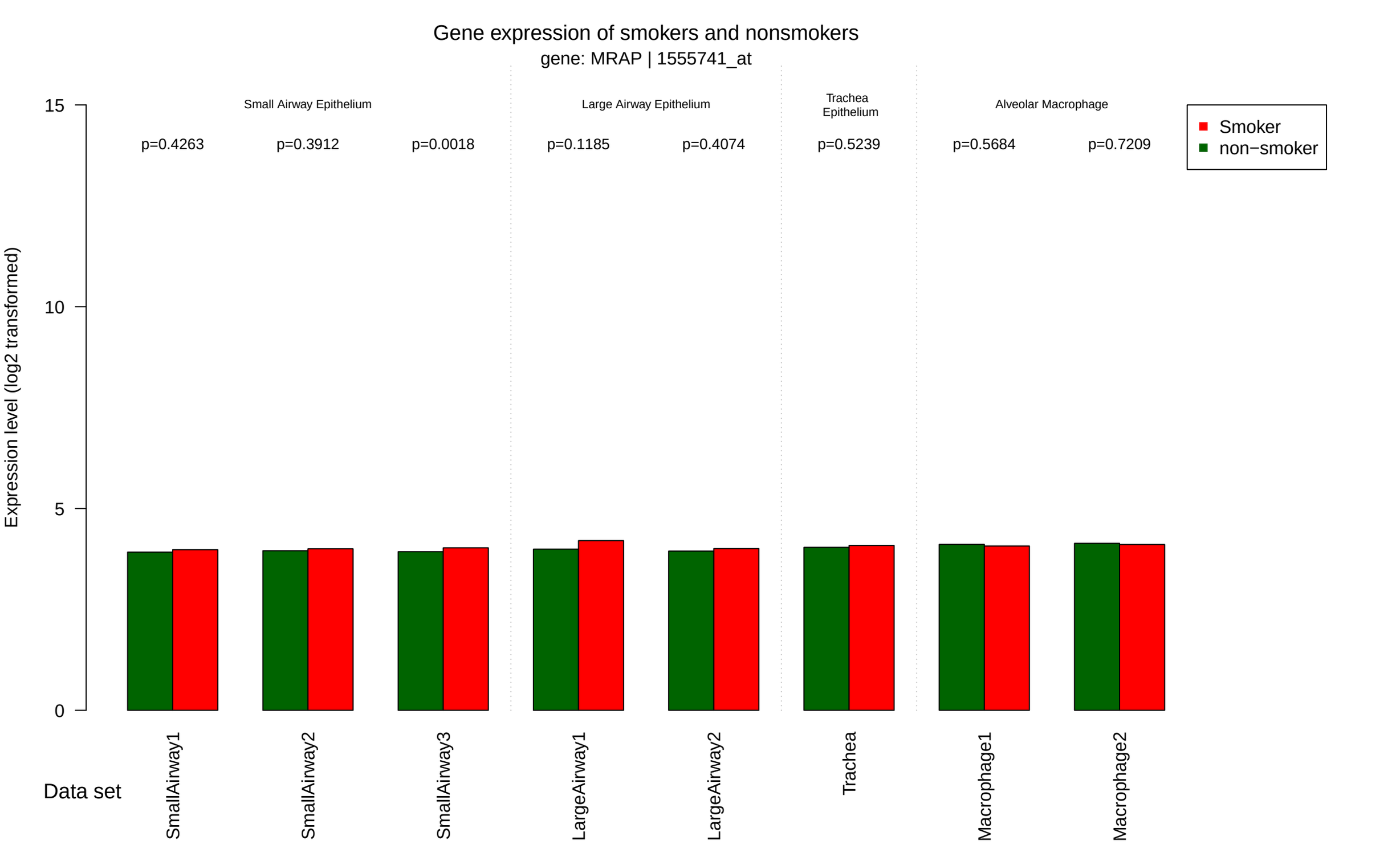
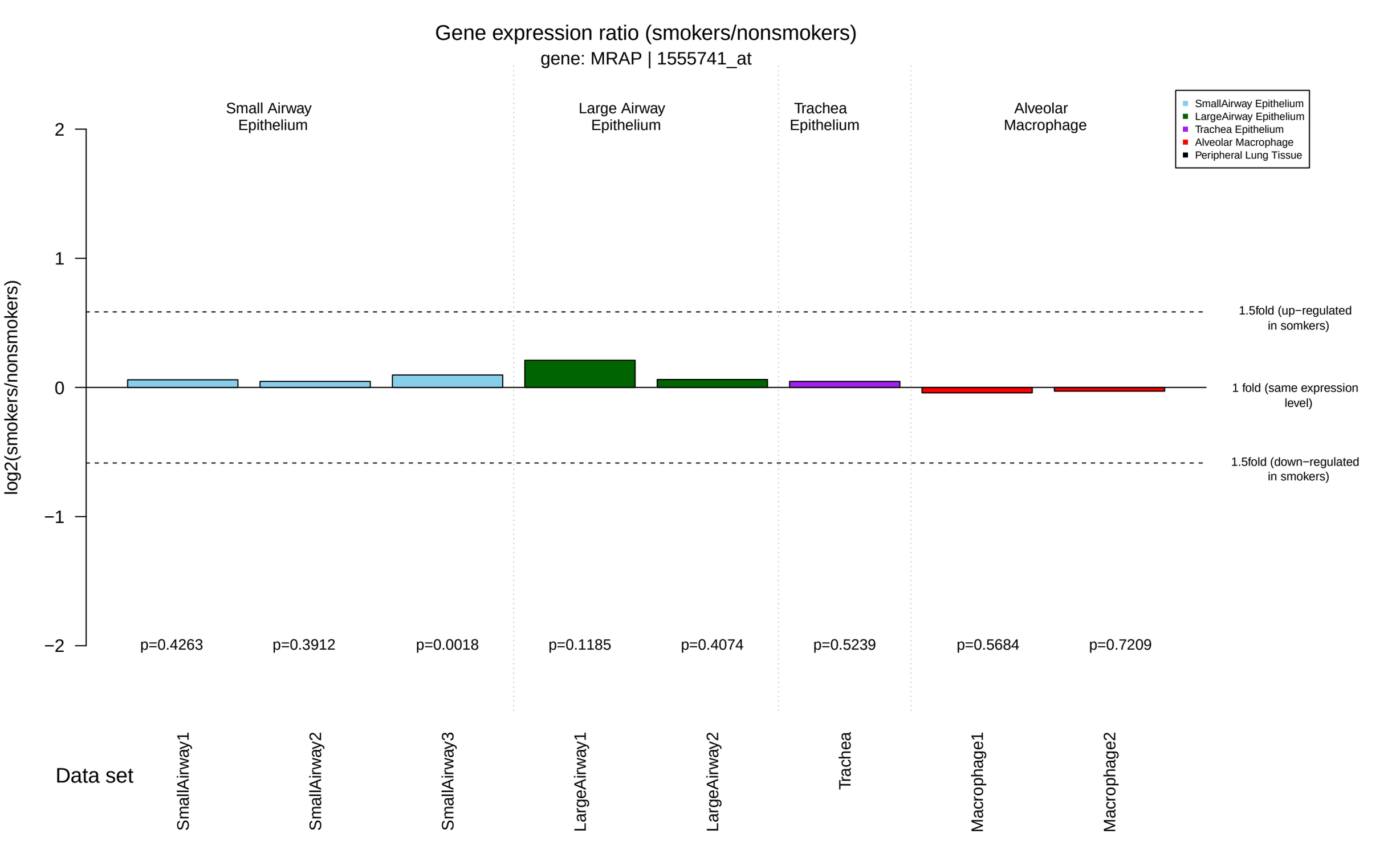
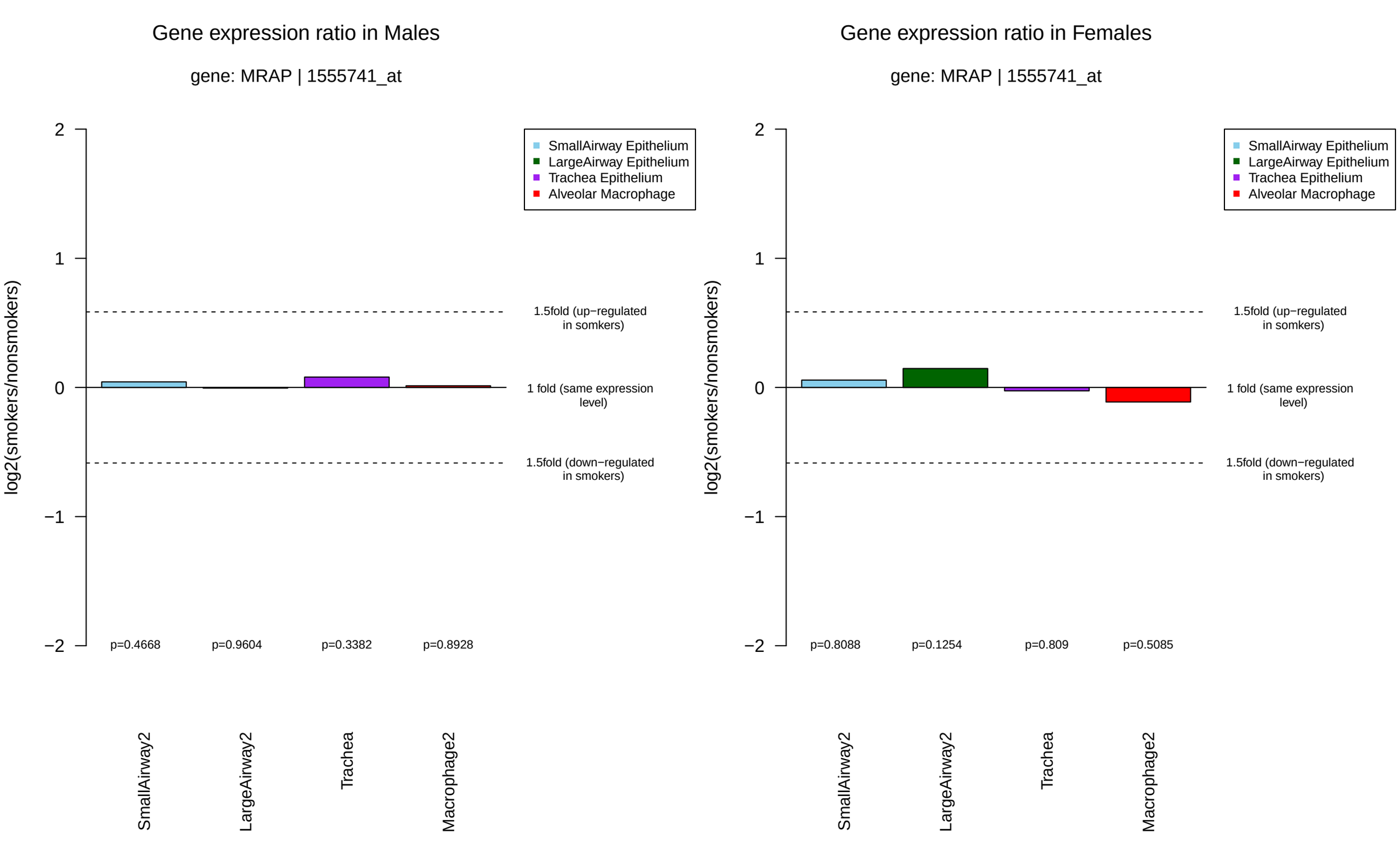
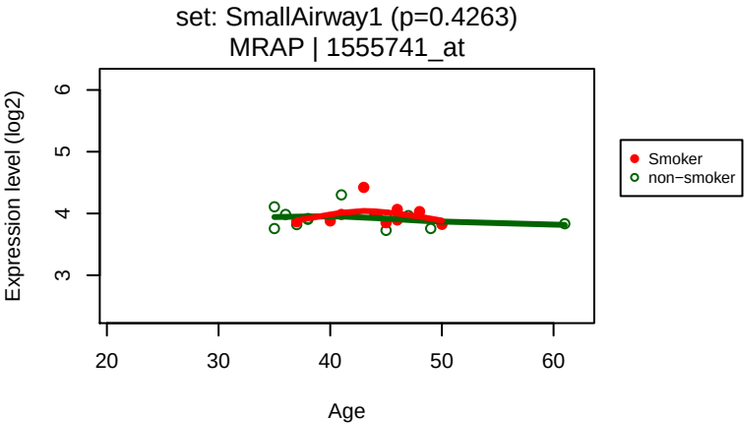
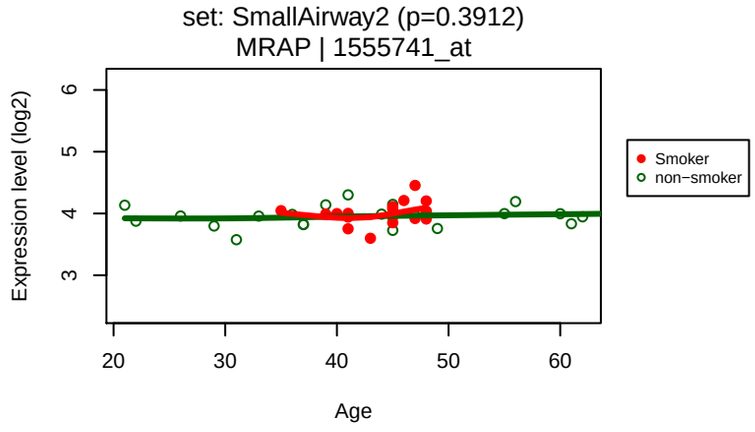
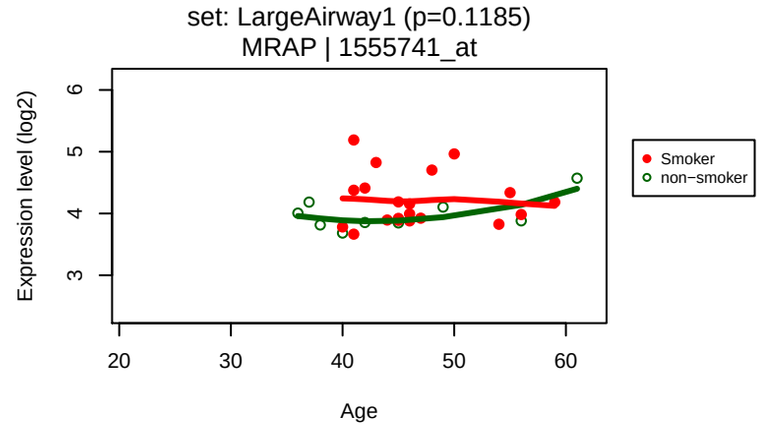

Gene expression of smokers and nonsmokers
gene: MRAP | 1555741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4263
p=0.3912
p=0.0018
p=0.1185
p=0.4074
p=0.5239
p=0.5684
p=0.7209
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

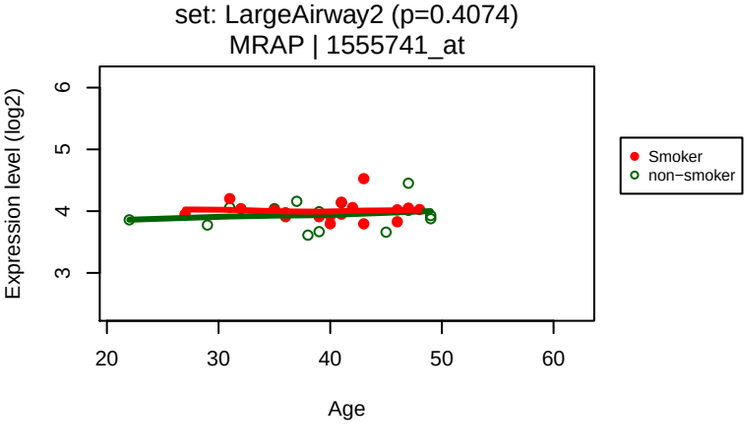

Gene expression ratio (smokers/nonsmokers)
gene: MRAP | 1555741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4263
p=0.3912
p=0.0018
p=0.1185
p=0.4074
p=0.5239
p=0.5684
p=0.7209
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

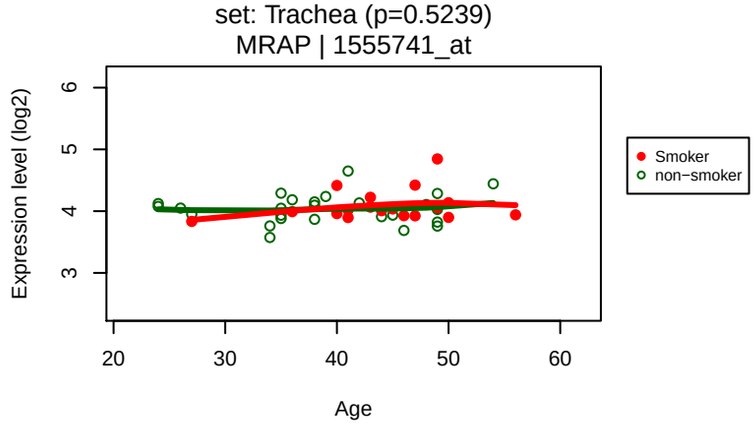

Gene expression ratio in Males
gene: MRAP | 1555741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4668
p=0.9604
p=0.3382
p=0.8928
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRAP | 1555741_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8088
p=0.1254
p=0.809
p=0.5085
SmallAirway2
LargeAirway2
Trachea
Macrophage2

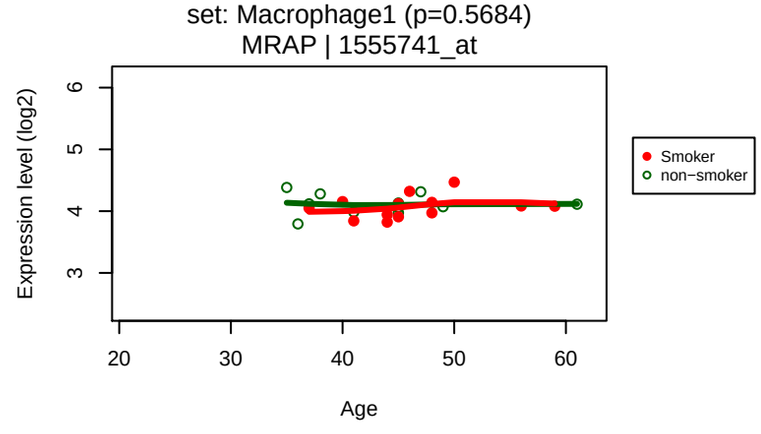

set: SmallAirway1 (p=0.4263)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker

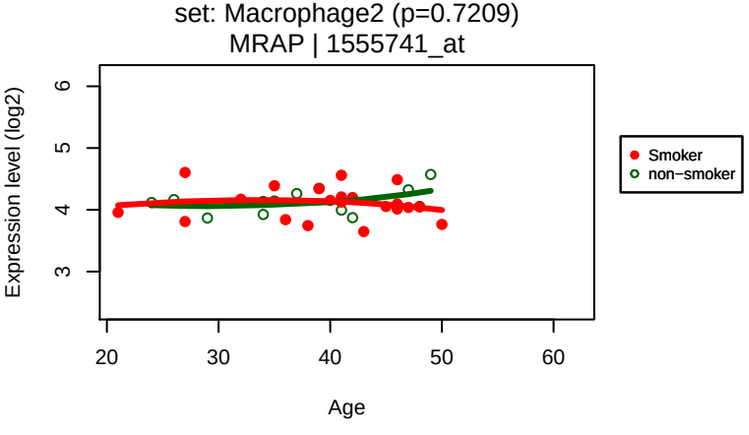

set: SmallAirway2 (p=0.3912)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker

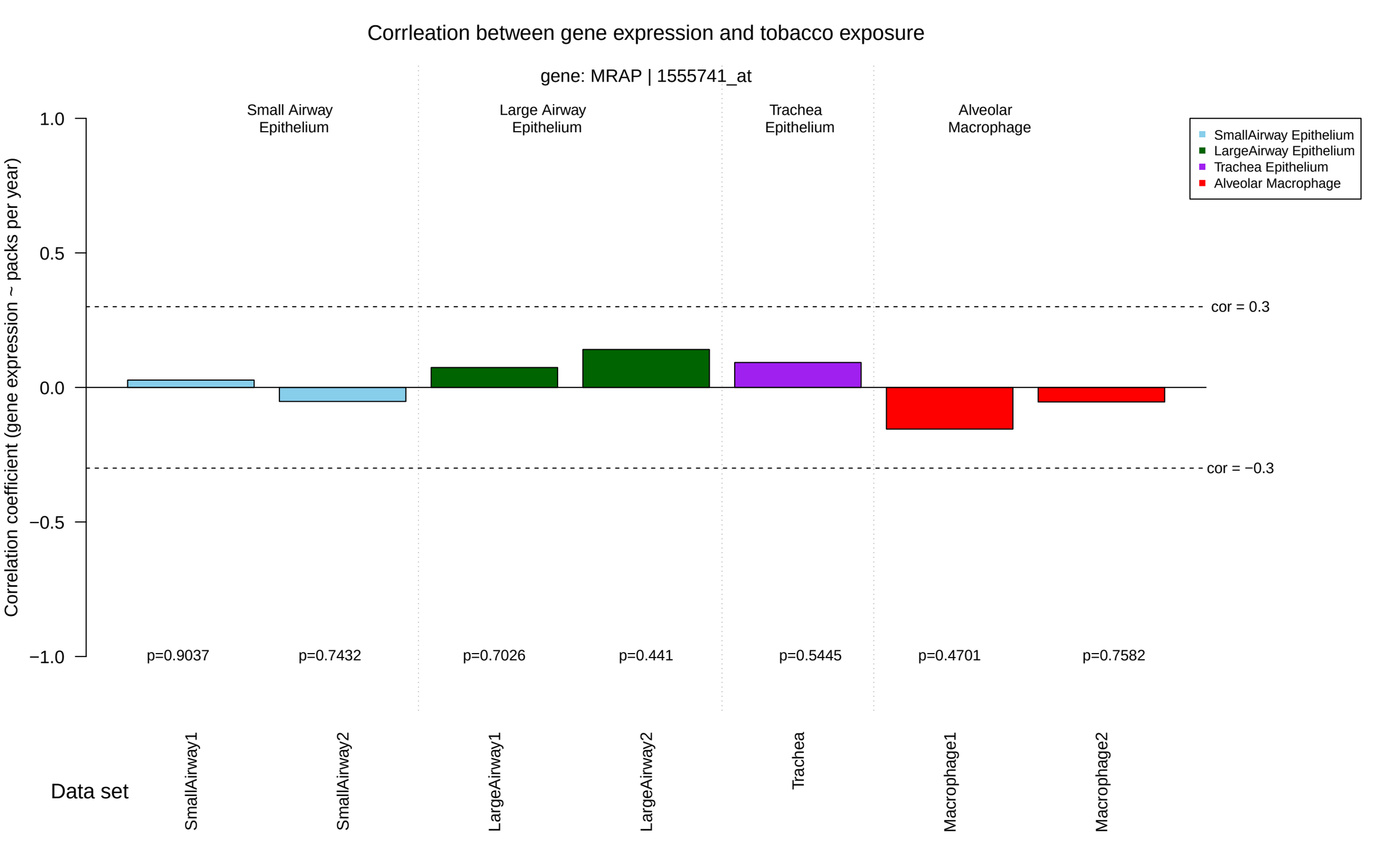

set: LargeAirway1 (p=0.1185)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4074)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5239)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5684)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7209)
MRAP | 1555741_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRAP | 1555741_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9037
p=0.7432
p=0.7026
p=0.441
p=0.5445
p=0.4701
p=0.7582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2